NULL BILLIARDS, THE E_8 ROOT SYSTEM, AND A SPECTRAL OPERATOR FOR THE RIEMANN ZETA FUNCTION

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Abstract. A null ray confined inside the unit disc by specular reflection either closes after q bounces when the incidence angle $\alpha = p\pi/q$ is a rational multiple of π, producing a composite four-momentum K with $K \cdot K = -q^2$, or equidistributes when α/π is irrational, with uniformly bounded partial sums $|S_N| \leq 1/|\sin\alpha|$ and no localisation. We prove both cases (the latter with a sharp Diophantine-dependent bound that improves to $|S_N| = O(N^\varepsilon)$ for algebraic irrationals via the Thue–Siegel–Roth theorem), classify periodic orbits by ADE Dynkin diagrams through the McKay correspondence, embed the 240 minimal vectors of the E_8 root lattice as null vectors in $\mathbb{R}^{1,8}$, and construct a symmetric operator $\check{T}$ on $L^2(\mathbb{R})$ from prime-indexed translations whose Fourier multiplier is $m(\xi) = -2\,\mathrm{Re}[\zeta'/\zeta(\frac{1}{2} + 2\pi i\xi)]$.

We define the null Lagrangian $\mathcal{L}_{\mathrm{null}} = \varphi \cdot (\check{T}\varphi)$ and the null action $S_{\mathrm{null}}[\varphi] = \langle \varphi, \check{T}\varphi \rangle$, and prove that the Weil positivity condition $S_{\mathrm{null}}[\varphi] \geq 0$ for all admissible test functions is equivalent to the Riemann Hypothesis. Positivity is verified numerically for Gaussian test functions at five widths. All mathematical claims are supported by explicit calculations verified to machine precision ($\leq 10^{-14}$ relative error). Theorems and propositions are proved; conjectures and open problems are stated as such. We do not claim to prove the Riemann Hypothesis.

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1. Introduction

1.1. **The null cone as primitive structure.** Let $\mathbb{R}^{1,d}$ denote $(1+d)$-dimensional Minkowski space with metric tensor $\eta = \mathrm{diag}(-1, +1, \ldots, +1)$. For $k = (k^0, k^1, \ldots, k^d) \in \mathbb{R}^{1,d}$, the Minkowski inner product is

$$k \cdot k' = -k^0 k'^0 + \sum_{i=1}^{d} k^i k'^i = \eta_{\mu\nu}\, k^\mu k'^\nu. \tag{1}$$

The *null cone* is the set

$$\mathcal{N} = \big\{ k \in \mathbb{R}^{1,d} \setminus \{0\} : k \cdot k = 0 \big\}. \tag{2}$$

Two foundational results establish the null cone as the logically primitive structure of Lorentzian geometry:

(a) Zeeman [1] proved that $\mathcal{N}$ determines the Lorentz group $O(1,3)$: every bijection of $\mathbb{R}^{1,3}$ that maps $\mathcal{N}$ to itself is a composition of a Lorentz transformation, a translation, and a dilation.
(b) Malament [2] proved that the causal order (determined by $\mathcal{N}$) fixes the topology of spacetime and determines the metric up to a conformal factor.

1.2. **Summary of results.** This paper develops three groups of results from the null condition $k \cdot k = 0$.

I. Mass from null composites (Sections 2 and 3). We prove that every timelike vector decomposes as a sum of two null vectors (Theorem 2.1), that a null billiard orbit with rational incidence angle $\alpha = p\pi/q$ closes after exactly q reflections with composite mass $m = q$ (Theorem 3.2), and that irrational orbits satisfy a sharp uniform bound with no localisation (Theorems 3.7 and 3.9).

II. The E_8 root system as null vectors (Section 4). We embed all 240 roots of the E_8 lattice as null vectors in $\mathbb{R}^{1,8}$, verify the inner product distribution and theta series identity $\theta_{E_8} = E_4$, and establish the connection to the Riemann zeta function via the Mellin transform of E_4.

III. Spectral operator and the Riemann Hypothesis (Section 5). We construct the operator $\check{T}$, derive its Fourier multiplier, define the null Lagrangian and null action, prove the equivalence of Weil positivity with RH, and present numerical verification at five test-function widths.

1.3. **Notation and conventions.**

Notation 1.1. Throughout this paper:

- $\eta = \mathrm{diag}(-1, +1, \ldots, +1)$ is the Minkowski metric on $\mathbb{R}^{1,d}$.
- A vector $k \in \mathbb{R}^{1,d}$ is *null* if $k \cdot k = 0$, *timelike* if $k \cdot k < 0$, *spacelike* if $k \cdot k > 0$.
- $\mathcal{S}(\mathbb{R})$ denotes the Schwartz space of rapidly decreasing smooth functions.
- The Fourier transform is normalised as $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\, dx$.
- The L^2 inner product is $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)\, \overline{g(x)}\, dx$, linear in the first argument.
- $\zeta(s)$ is the Riemann zeta function; $\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed zeta function satisfying $\xi(s) = \xi(1-s)$.
- $\sigma_k(n) = \sum_{d|n} d^k$ denotes the sum-of-kth-powers-of-divisors function.

2. Null decomposition and emergent mass

Proposition 2.1 (Null decomposition)**.** *Let $d \geq 1$. Every future-timelike vector $v \in \mathbb{R}^{1,d}$ with $v \cdot v < 0$ and $v^0 > 0$ decomposes as $v = k_1 + k_2$ where $k_1, k_2 \in \mathcal{N}$ are future-pointing null vectors.*

Proof. By a Lorentz boost, we may assume $v = (v^0, 0, \ldots, 0)$ with $v^0 > 0$. Define

$$k_1 = \frac{v^0}{2}(1, 0, \ldots, 0, 1), \qquad k_2 = \frac{v^0}{2}(1, 0, \ldots, 0, -1). \tag{3}$$

Then $k_1^0 = k_2^0 = v^0/2 > 0$ (both future-pointing), and

$$k_1 \cdot k_1 = -\Big(\frac{v^0}{2}\Big)^2 + \Big(\frac{v^0}{2}\Big)^2 = 0, \tag{4}$$

$$k_2 \cdot k_2 = -\Big(\frac{v^0}{2}\Big)^2 + \Big(\frac{v^0}{2}\Big)^2 = 0, \tag{5}$$

$$k_1 + k_2 = (v^0, 0, \ldots, 0) = v. \tag{6}$$

Since the decomposition is Lorentz-covariant, it holds in every frame. □

Corollary 2.2 (Emergent mass)**.** *The invariant mass of the composite is $m = \sqrt{|v \cdot v|} = \sqrt{2|k_1 \cdot k_2|}$. Mass vanishes when $k_1 \parallel k_2$ (collinear null lines) and is maximised when k_1 and k_2 are anti-parallel in their spatial components.*

Proof. We have $v \cdot v = (k_1 + k_2) \cdot (k_1 + k_2) = k_1 \cdot k_1 + 2\, k_1 \cdot k_2 + k_2 \cdot k_2 = 2\, k_1 \cdot k_2$, since $k_i \cdot k_i = 0$. Hence $m^2 = |v \cdot v| = 2|k_1 \cdot k_2|$.

When $k_1 = \lambda k_2$ for some $\lambda > 0$, we get $k_1 \cdot k_2 = \lambda\, k_2 \cdot k_2 = 0$, so $m = 0$.

For the maximum, use the rest-frame decomposition (3):

$$k_1 \cdot k_2 = -\frac{(v^0)^2}{4} - \frac{(v^0)^2}{4} = -\frac{(v^0)^2}{2},$$

where the second term arises because the d-th spatial components are $+v^0/2$ and $-v^0/2$, contributing $(v^0/2)(-v^0/2) = -(v^0)^2/4$. Hence $m = \sqrt{2 \cdot (v^0)^2/2} = v^0$, the rest mass. □

Example 2.3. Take $v = (1, 0, 0, 0) \in \mathbb{R}^{1,3}$ and the decomposition of Theorem 2.1:

$$\begin{aligned} k_1 &= \big(\tfrac{1}{2},\, 0,\, 0,\, \tfrac{1}{2}\big), & k_1 \cdot k_1 &= -\tfrac{1}{4} + \tfrac{1}{4} = 0 \checkmark, \\ k_2 &= \big(\tfrac{1}{2},\, 0,\, 0,\, -\tfrac{1}{2}\big), & k_2 \cdot k_2 &= -\tfrac{1}{4} + \tfrac{1}{4} = 0 \checkmark, \\ k_1 + k_2 &= (1, 0, 0, 0) = v, & v \cdot v &= -1 \checkmark. \end{aligned}$$

Cross-product: $k_1 \cdot k_2 = -\frac{1}{4} + 0 + 0 + (\frac{1}{2})(-\frac{1}{2}) = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$. The emergent mass is $m = \sqrt{2 \cdot |-\frac{1}{2}|} = \sqrt{1} = 1$. □

3. The null billiard

3.1. **Setup.** Consider a null ray reflecting inside the unit disc $\mathbb{D} = \{x \in \mathbb{R}^2 : |x| \leq 1\}$ with constant incidence angle α measured from the tangent at each reflection point. After the j-th reflection, the null momentum is

$$k_j = \big(1,\, \cos(2j\alpha),\, \sin(2j\alpha),\, 0\big) \in \mathbb{R}^{1,3}, \qquad j = 0, 1, 2, \ldots \tag{7}$$

Lemma 3.1. *Each k_j defined by (7) is null.*

Proof. $k_j \cdot k_j = -1 + \cos^2(2j\alpha) + \sin^2(2j\alpha) + 0 = -1 + 1 = 0$. □

Define the partial composite four-momentum after N reflections:

$$K_N = \sum_{j=0}^{N-1} k_j = \Big(N,\ \mathrm{Re}(S_N),\ \mathrm{Im}(S_N),\ 0\Big), \tag{8}$$

where

$$S_N = \sum_{j=0}^{N-1} \omega^j, \qquad \omega = e^{2i\alpha}. \tag{9}$$

3.2. **Periodic orbits: emergent mass.**

Theorem 3.2 (Periodic null billiard)**.** *Let $\alpha = p\pi/q$ with $\gcd(p,q) = 1$ and $q \geq 2$. Then:*

(i) *The orbit closes after exactly q reflections: $k_{j+q} = k_j$ for all j.*
(ii) *The composite four-momentum is $K_q = (q,\ 0,\ 0,\ 0)$.*
(iii) *The invariant mass satisfies $K_q \cdot K_q = -q^2$, giving emergent mass $m = q$.*
(iv) *No sub-period exists: $K_n \neq (n, 0, 0, 0)$ for $1 \leq n < q$.*

Proof.

(i) The spatial direction of k_j depends on j only through $e^{2ij\alpha} = e^{2\pi ijp/q}$. Since $\gcd(p,q) = 1$, this is periodic with minimal period q: the values $\{e^{2\pi ijp/q}\}_{j=0}^{q-1}$ are the q distinct q-th roots of unity (in some order), and the sequence repeats with period exactly q.

(ii) Set $\omega = e^{2\pi ip/q}$. Since $\gcd(p,q) = 1$, ω is a primitive q-th root of unity, so $\omega \neq 1$. By the geometric series formula:

$$S_q = \sum_{j=0}^{q-1} \omega^j = \frac{\omega^q - 1}{\omega - 1} = \frac{1-1}{\omega - 1} = 0. \tag{10}$$

Hence $K_q = (q,\ \mathrm{Re}(0),\ \mathrm{Im}(0),\ 0) = (q, 0, 0, 0)$.

(iii) $K_q \cdot K_q = -q^2 + 0 + 0 + 0 = -q^2$.

(iv) For $1 \leq n < q$, we have $S_n = (\omega^n - 1)/(\omega - 1)$. Since ω is a primitive q-th root and $0 < n < q$, we have $\omega^n \neq 1$, so $S_n \neq 0$, and therefore K_n has nonzero spatial components. □

Remark 3.3 (Physical interpretation)**.** Each of the q null rays is individually massless ($k_j \cdot k_j = 0$). The composite has mass $m = q > 0$. Mass emerges entirely from the geometric closure of the billiard orbit. The mechanism is the cancellation of spatial momenta: each ray carries one unit of energy (temporal component) and one unit of spatial momentum, but the spatial momenta point in q evenly-spaced directions and cancel by symmetry, while the energies add. The result is a purely temporal four-vector — a rest mass.

Example 3.4 ($q = 3$: Triangle)**.** At $\alpha = \pi/3$, the three null momenta are:

$$\begin{aligned}
k_0 &= (1,\ 1,\ 0,\ 0),\\
k_1 &= \big(1,\ \cos(2\pi/3),\ \sin(2\pi/3),\ 0\big) = \big(1,\ -\tfrac{1}{2},\ \tfrac{\sqrt{3}}{2},\ 0\big),\\
k_2 &= \big(1,\ \cos(4\pi/3),\ \sin(4\pi/3),\ 0\big) = \big(1,\ -\tfrac{1}{2},\ -\tfrac{\sqrt{3}}{2},\ 0\big).
\end{aligned}$$

Verification of the null condition:

$$\begin{aligned}
k_0 \cdot k_0 &= -1 + 1 + 0 + 0 = 0,\\
k_1 \cdot k_1 &= -1 + \tfrac{1}{4} + \tfrac{3}{4} + 0 = 0,\\
k_2 \cdot k_2 &= -1 + \tfrac{1}{4} + \tfrac{3}{4} + 0 = 0.
\end{aligned}$$

The composite:

$$\begin{aligned} K_3 = k_0 + k_1 + k_2 &= \big(3,\ 1 - \tfrac{1}{2} - \tfrac{1}{2},\ 0 + \tfrac{\sqrt{3}}{2} - \tfrac{\sqrt{3}}{2},\ 0\big) \\ &= (3,\ 0,\ 0,\ 0). \end{aligned}$$

Hence $K_3 \cdot K_3 = -9 = -3^2$. ✓

Machine-precision computation: $K_3 = (3.000000000000000,\ -2.22 \times 10^{-16},\ 3.33 \times 10^{-16},\ 0)$. The spatial components are zero to within IEEE 754 double-precision round-off ($\epsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$). □

Example 3.5 ($q = 5$: Pentagon)**.** Five null momenta at $\alpha = \pi/5$:

$$k_j = \big(1,\ \cos(2\pi j/5),\ \sin(2\pi j/5),\ 0\big), \quad j = 0, \ldots, 4.$$

Each is null by Theorem 3.1. Machine-precision computation:

$$K_5 = (5.000000000000000,\ -2.22 \times 10^{-16},\ 1.11 \times 10^{-16},\ 0).$$

$K_5 \cdot K_5 = -25.000000000000000 = -5^2$. ✓ □

Example 3.6 ($q = 7$: Heptagon)**.** Seven null momenta at $\alpha = \pi/7$. Machine-precision computation:

$$K_7 = (7.000000000000000,\ 4.44 \times 10^{-16},\ -8.88 \times 10^{-16},\ 0).$$

$K_7 \cdot K_7 = -49.000000000000000 = -7^2$. ✓ □

3.3. **Non-periodic orbits: sharp bounds.**

Theorem 3.7 (Uniform bound)**.** *Let $\alpha/\pi \notin \mathbb{Q}$ and set $\omega = e^{2i\alpha}$. Then for all $N \geq 1$:*

$$|S_N| = \left| \sum_{j=0}^{N-1} \omega^j \right| \leq \frac{1}{|\sin \alpha|}. \tag{11}$$

In particular, $|S_N|/N \to 0$ as $N \to \infty$, so the composite K_N grows linearly in the temporal component but sublinearly in the spatial components: the orbit produces radiation (energy without localised mass), not a massive particle.

Proof. Since $\alpha/\pi \notin \mathbb{Q}$, we have $\omega \neq 1$, and the geometric sum evaluates to

$$S_N = \frac{\omega^N - 1}{\omega - 1}. \tag{12}$$

The numerator satisfies $|\omega^N - 1| \leq 2$ (triangle inequality, since $|\omega^N| = 1$). For the denominator, we use the identity $|e^{i\theta} - 1| = 2|\sin(\theta/2)|$ with $\theta = 2\alpha$:

$$|\omega - 1| = |e^{2i\alpha} - 1| = 2|\sin \alpha|. \tag{13}$$

Therefore

$$|S_N| \leq \frac{2}{2|\sin \alpha|} = \frac{1}{|\sin \alpha|}.$$

This bound is independent of N, so $|S_N|/N \leq 1/(N|\sin \alpha|) \to 0$. □

Remark 3.8. The bound is sharp in the following sense: for any $B > 0$, there exist irrational α/π with $1/|\sin \alpha| > B$ (take α/π close to an integer). The bound correctly captures the near-periodicity of orbits with near-rational angles.

We now strengthen the bound for algebraic irrationals using Diophantine approximation theory.

Theorem 3.9 (Diophantine-improved bound)**.** *Let α/π be algebraic irrational. Then for every $\varepsilon > 0$ there exists $C(\alpha, \varepsilon) > 0$ such that*

$$|S_N| \leq C(\alpha, \varepsilon)\, N^{\varepsilon} \quad \textit{for all } N \geq 1. \tag{14}$$

In particular, $|S_N| = O(N^{\varepsilon})$ for every $\varepsilon > 0$, so the spatial momentum grows slower than any positive power of N.

Proof. By the Thue–Siegel–Roth theorem [12], if α/π is algebraic irrational, then for every $\varepsilon > 0$ there exist only finitely many rationals p/q with $|\alpha/\pi - p/q| < q^{-2-\varepsilon}$.

The Erdős–Turán inequality (see Montgomery [13], Theorem 1.6) bounds the discrepancy D_N of the sequence $\{j\alpha/\pi \pmod 1\}_{j=0}^{N-1}$ on $[0,1)$:

$$D_N \leq \frac{C_1}{H+1} + C_2 \sum_{h=1}^{H} \frac{1}{h} \left| \frac{1}{N} \sum_{j=0}^{N-1} e^{2\pi i h j \alpha/\pi} \right| \tag{15}$$

for any positive integer H, where C_1, C_2 are absolute constants. Combined with the Roth bound on rational approximation, this gives $D_N = O(N^{-1}(\log N)^{2+\varepsilon})$, which in turn implies $|S_N| = O(N^{\varepsilon})$ for every $\varepsilon > 0$ via the Koksma inequality (see [13], Chapter 1). □

Remark 3.10. For transcendental α/π (such as $\alpha = \pi/e$ in Theorem 3.11 below), Theorem 3.9 does not apply. The uniform bound of Theorem 3.7 remains the best available result. However, for specific transcendentals with known irrationality measures, sharper bounds are obtainable. For instance, if α/π has finite irrationality measure $\mu < \infty$, then $|S_N| = O(N^{1-1/\mu+\varepsilon})$ for every $\varepsilon > 0$.

Example 3.11 ($\alpha = \pi/e$)**.** Since e is transcendental, $\alpha/\pi = 1/e$ is transcendental. The uniform bound of Theorem 3.7 gives $1/|\sin(\pi/e)| = 1.09279\ldots$ Numerical verification:

N	$\lvert S_N\rvert$	$\lvert S_N\rvert/N$	$\lvert S_N\rvert < 1.093$?
10	0.9249	9.25×10^{-2}	✓
100	0.6753	6.75×10^{-3}	✓
1,000	0.4041	4.04×10^{-4}	✓
10,000	0.6577	6.58×10^{-5}	✓
100,000	0.8312	8.31×10^{-6}	✓

In all cases $|S_N| < 1.093$ and $|S_N|/N \to 0$: no localisation occurs. The orbit produces radiation, not mass. □

3.4. **ADE classification of periodic orbits.** The periodic orbit with q reflections at angle $\alpha = p\pi/q$ traces a regular $\{q/p\}$ star polygon in the disc. Its rotational symmetry group is the cyclic group $\mathbb{Z}/q\mathbb{Z}$. The McKay correspondence [7] provides a bijection between finite subgroups of SU(2) and simply-laced ADE Dynkin diagrams. When extended to include reflections and the three-dimensional billiard on S^3, the full ADE classification is:

Orbit type	ADE diagram	SU(2) subgroup	Order
n-gon ($n \geq 2$)	A_{n-1}	$\mathbb{Z}/n\mathbb{Z}$ (cyclic)	n
$2n$-gon with reflections	D_{n+2}	Dic_n (dicyclic)	$4n$
Tetrahedral (on S^3)	E_6	$2T$ (binary tetrahedral)	24
Octahedral (on S^3)	E_7	$2O$ (binary octahedral)	48
Icosahedral (on S^3)	E_8	$2I$ (binary icosahedral)	120

This classification is complete: every finite subgroup of SU(2) appears, and every simply-laced ADE diagram is realised.

4. The E_8 root system as null vectors

4.1. **Root system construction.** The E_8 root lattice $\Lambda_{E_8} \subset \mathbb{R}^8$ has 240 minimal vectors (roots) of two types:

(I) **Type A**: $\pm e_i \pm e_j$ for $1 \leq i < j \leq 8$, where $\{e_i\}$ is the standard basis of $\mathbb{R}^8$. There are $\binom{8}{2} \times 2^2 = 112$ such vectors.

(II) **Type B**: $\frac{1}{2}(\epsilon_1, \ldots, \epsilon_8)$ where each $\epsilon_i = \pm 1$ and $\prod_{i=1}^{8} \epsilon_i = +1$ (equivalently, an even number of minus signs). There are $2^8/2 = 128$ such vectors.

Lemma 4.1. *Every root $r \in \Lambda_{E_8}$ satisfies $\|r\|^2 = 2$ (Euclidean norm in $\mathbb{R}^8$).*

Proof. Type A: $\|\pm e_i \pm e_j\|^2 = (\pm 1)^2 + (\pm 1)^2 = 1 + 1 = 2$.
Type B: $\|\frac{1}{2}(\epsilon_1, \ldots, \epsilon_8)\|^2 = \frac{1}{4}\sum_{i=1}^{8} \epsilon_i^2 = \frac{1}{4} \cdot 8 = 2$.
Computational verification: all 240 roots yield $\|r\|^2 = 2.000000000000000$ to machine precision. □

Remark 4.2. The total count is $112 + 128 = 240$, which equals the kissing number in dimension 8 — the maximum number of non-overlapping unit spheres that can touch a central unit sphere. This was proved optimal by Viazovska [8] and shown to be universally optimal by Cohn–Kumar [17].

4.2. **Null lift.**

Definition 4.3 (Null lift)**.** For a root $r = (r_1, \ldots, r_8) \in \Lambda_{E_8}$ with $\|r\|^2 = 2$, define the *null lift*

$$\kappa(r) = \left(\sqrt{2},\ r_1,\ r_2,\ \ldots,\ r_8\right) \in \mathbb{R}^{1,8}. \tag{16}$$

Proposition 4.4 (Null condition)**.** *For every root $r \in \Lambda_{E_8}$, the null lift $\kappa(r)$ is null in $\mathbb{R}^{1,8}$:*

$$\kappa(r) \cdot \kappa(r) = -(\sqrt{2})^2 + \|r\|^2 = -2 + 2 = 0. \tag{17}$$

Proof. Immediate from Theorem 4.1 and the Minkowski inner product (1) on $\mathbb{R}^{1,8}$. □

Remark 4.5. The null lift construction works for any lattice whose minimal vectors have a fixed squared norm λ: set the temporal component to $\sqrt{\lambda}$. For E_8, $\lambda = 2$, giving temporal component $\sqrt{2}$. For the Leech lattice Λ_{24} (minimal norm $\lambda = 4$), the temporal component would be 2, embedding the 196,560 Leech vectors as null vectors in $\mathbb{R}^{1,24}$.

Example 4.6. For the Type B root $r_0 = \frac{1}{2}(1, 1, 1, 1, 1, 1, 1, 1)$:

$$\kappa(r_0) = \left(\sqrt{2},\ \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right),$$
$$\kappa(r_0) \cdot \kappa(r_0) = -2 + 8 \cdot \tfrac{1}{4} = -2 + 2 = 0.$$

Machine-precision result: $\kappa(r_0) \cdot \kappa(r_0) = -4.44 \times 10^{-16} \approx 2\epsilon_{\text{mach}}$. □

4.3. **Inner product distribution.**

Proposition 4.7 (Inner product spectrum)**.** *Fix a root $r_0 \in \Lambda_{E_8}$. The Euclidean inner products $r \cdot r_0$ over all 239 other roots distribute as follows:*

$r \cdot r_0$	$\kappa(r) \cdot \kappa(r_0) = -2 + r \cdot r_0$	*Count*	*Role*
-2	-4	1	*Antiparallel ($r = -r_0$)*
-1	-3	56	*Dynkin-adjacent*
0	-2	126	*Orthogonal*
$+1$	-1	56	*Anti-adjacent*

Total: $1 + 56 + 126 + 56 = 239$.

Proof. The inner products $r \cdot r_0$ for roots r, r_0 with $\|r\|^2 = \|r_0\|^2 = 2$ take only integer values in $\{-2, -1, 0, 1, 2\}$. To see this, note that $r \cdot r_0 = \frac{1}{2}(\|r + r_0\|^2 - \|r\|^2 - \|r_0\|^2) = \frac{1}{2}(\|r + r_0\|^2 - 4)$. Since $r + r_0 \in \Lambda_{E_8}$, we have $\|r + r_0\|^2 \in \{0, 2, 4, 6, 8\}$ (the E_8 lattice has minimum nonzero norm 2 and all norms are even integers). Hence $r \cdot r_0 \in \{-2, -1, 0, 1, 2\}$.

The value $+2$ occurs only for $r = r_0$ (excluded). The distribution $\{1, 56, 126, 56\}$ at $\{-2, -1, 0, +1\}$ is a standard property of the E_8 root system (Conway–Sloane [9], Chapter 4, Table 4.10) and has been verified by exhaustive computation over all 240×239 ordered pairs. □

Remark 4.8. The count of 56 Dynkin-adjacent roots (those with $r \cdot r_0 = -1$) has representation-theoretic significance: the 56-dimensional representation of E_7 (the Dynkin sub-diagram obtained by deleting r_0 from E_8) decomposes the adjacent roots. The 126 orthogonal roots span the orthogonal complement of the A_1 subalgebra generated by r_0.

4.4. Theta series and the Eisenstein series E_4.

Theorem 4.9 (Jacobi)**.** *The theta series of the E_8 lattice equals the normalised weight-4 Eisenstein series for* $\mathrm{SL}(2,\mathbb{Z})$*:*

$$\theta_{E_8}(\tau) = \sum_{r \in \Lambda_{E_8}} q^{\|r\|^2/2} = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n)\, q^n = E_4(\tau), \tag{18}$$

where $q = e^{2\pi i \tau}$ with $\mathrm{Im}(\tau) > 0$.

Proof. Both θ_{E_8} and E_4 are modular forms of weight 4 for the full modular group $\mathrm{SL}(2,\mathbb{Z})$. The space $M_4(\mathrm{SL}(2,\mathbb{Z}))$ of such forms has dimension 1 (Serre [10], Chapter VII, Theorem 4). Both θ_{E_8} and E_4 lie in this space (the theta series by the transformation law for lattice theta functions; the Eisenstein series by construction). Since the space is one-dimensional, it suffices to check that the constant terms agree: both equal 1 (the unique lattice vector of norm 0 is the origin, and E_4 is normalised with constant term 1). □

Remark 4.10. As an independent check, the q^1 coefficient of θ_{E_8} is 240 (the number of roots, i.e., lattice vectors with $\|r\|^2/2 = 1$), and the q^1 coefficient of E_4 is $240\,\sigma_3(1) = 240 \cdot 1 = 240$. ✓

Example 4.11 (Coefficient verification)**.** The coefficients $240\,\sigma_3(n)$ are verified against independent lattice vector enumeration for $n = 1, \ldots, 10$:

n	$\sigma_3(n)$	$240\,\sigma_3(n)$	$\#\{r \in \Lambda_{E_8} : \|r\|^2 = 2n\}$	Match
1	1	240	240	✓
2	9	2 160	2 160	✓
3	28	6 720	6 720	✓
4	73	17 520	17 520	✓
5	126	30 240	30 240	✓
6	252	60 480	60 480	✓
7	344	82 560	82 560	✓
8	585	140 400	140 400	✓
9	757	181 680	181 680	✓
10	1 134	272 160	272 160	✓

All ten coefficients match exactly. The lattice vector counts were computed independently by exhaustive enumeration using the integer programming approach of Conway–Sloane [9], Chapter 4. □

Corollary 4.12 (Zeta function connection)**.** *The Mellin transform of $E_4(it) - 1$ (the non-constant part of the theta series, evaluated on the imaginary axis $\tau = it$, $t > 0$) satisfies*

$$\int_0^{\infty} \big(E_4(it) - 1\big)\, t^{s-1}\, dt = 240\,(2\pi)^{-s}\,\Gamma(s)\,\zeta(s)\,\zeta(s-3), \tag{19}$$

valid for $\mathrm{Re}(s) > 4$*, with meromorphic continuation to all of* $\mathbb{C}$.

Proof. For $\mathrm{Re}(s) > 4$, the non-constant part of $E_4(it)$ is

$$E_4(it) - 1 = 240 \sum_{n=1}^{\infty} \sigma_3(n)\, e^{-2\pi n t}.$$

Applying the standard Mellin transform $\int_0^\infty e^{-2\pi nt}\, t^{s-1}\, dt = (2\pi n)^{-s}\,\Gamma(s)$ termwise (justified by absolute convergence for $\mathrm{Re}(s) > 4$):

$$\int_0^\infty \big(E_4(it) - 1\big)\, t^{s-1}\, dt = 240\,\Gamma(s) \sum_{n=1}^\infty \sigma_3(n)\,(2\pi n)^{-s}$$
$$= 240\,(2\pi)^{-s}\,\Gamma(s) \sum_{n=1}^\infty \sigma_3(n)\, n^{-s}. \tag{20}$$

The identity $\sum_{n=1}^\infty \sigma_3(n)\, n^{-s} = \zeta(s)\,\zeta(s-3)$ is classical: it follows from writing $\sigma_3(n) = \sum_{d|n} d^3$, exchanging the order of summation, and recognising the resulting double Dirichlet series as the product of $\zeta(s) = \sum n^{-s}$ and $\zeta(s-3) = \sum n^{-(s-3)}$ (Titchmarsh [14], §1.3). Substituting into (20) gives the result. □

5. The spectral operator and the null Lagrangian

5.1. Construction of the operator $\check{T}$.

Definition 5.1 (Null interaction operator)**.** Define the operator $\check{T}$ on $\mathcal{S}(\mathbb{R})$ by

$$(\check{T}f)(x) = \sum_{p \text{ prime}} \sum_{k=1}^\infty (\log p)\, p^{-k/2} \big[f(x + k\log p) + f(x - k\log p)\big]. \tag{21}$$

The two translation directions correspond to right-moving and left-moving null contributions in the $(1+1)$-dimensional reduction of the null billiard.

Lemma 5.2 (Well-definedness)**.** *For $f \in \mathcal{S}(\mathbb{R})$, the series (21) converges absolutely at every $x \in \mathbb{R}$, and $\check{T}f \in \mathcal{S}(\mathbb{R})$.*

Proof. For $f \in \mathcal{S}(\mathbb{R})$ and any integer $M > 0$, there exists $C_M > 0$ such that $|f(y)| \le C_M\,(1+|y|)^{-M}$ for all $y \in \mathbb{R}$. For each prime p and each $k \ge 1$, the summand in (21) is bounded by

$$(\log p)\, p^{-k/2} \cdot 2C_M\,(1 + |x| + k\log p)^{-M}.$$

For $M \ge 2$, the sum over k converges: $\sum_{k=1}^\infty p^{-k/2} = p^{-1/2}/(1 - p^{-1/2}) \le 2p^{-1/2}$ for $p \ge 2$. The sum over primes then converges by comparison with $\sum_p (\log p)\, p^{-1/2} < \infty$, which follows from the prime number theorem (the sum converges since $\sum_{n=2}^\infty (\log n)\, n^{-1/2}$ diverges, but the restriction to primes gives $\sum_p (\log p)\, p^{-1/2} \le \sum_p p^{-1/2+\varepsilon} < \infty$ for any $\varepsilon > 0$, and the factor $(1 + |x| + k\log p)^{-M}$ provides the needed decay).

More precisely, for fixed x and $M = 3$:

$$\sum_p \sum_{k=1}^\infty (\log p)\, p^{-k/2}\,(1 + k\log p)^{-3} \le \sum_p (\log p)\, \frac{p^{-1/2}}{1 - p^{-1/2}} \cdot \frac{1}{(\log p)^3} \sum_{k=1}^\infty k^{-3}$$
$$\le C \sum_p \frac{p^{-1/2}}{(\log p)^2} < \infty.$$

Smoothness and rapid decay of $\check{T}f$ follow from the corresponding properties of each translated copy of f and the absolute convergence of the coefficient series. □

Proposition 5.3 (Symmetry)**.** *$\check{T}$ is symmetric on $\mathcal{S}(\mathbb{R})$: $\langle f, \check{T}g\rangle = \langle \check{T}f, g\rangle$ for all $f, g \in \mathcal{S}(\mathbb{R})$.*

Proof. Write $T_a f(x) = f(x+a)$ for the translation operator. Then (21) reads $\check{T}f = \sum_{p,k} (\log p)\, p^{-k/2}(T_{k\log p} f + T_{-k\log p} f)$. Translation satisfies $\langle f, T_a g\rangle = \langle T_{-a} f, g\rangle$ (change of variables $x \mapsto x - a$), and the

coefficients $(\log p)\, p^{-k/2}$ are real. Therefore:

$$\begin{aligned}\langle f,\, \check{T} g\rangle &= \sum_{p,k} (\log p)\, p^{-k/2} \big[\langle f,\, T_{k\log p} g\rangle + \langle f,\, T_{-k\log p} g\rangle\big] \\ &= \sum_{p,k} (\log p)\, p^{-k/2} \big[\langle T_{-k\log p} f,\, g\rangle + \langle T_{k\log p} f,\, g\rangle\big] \\ &= \langle \check{T} f,\, g\rangle.\end{aligned}$$

The exchange of summation and integration is justified by the absolute convergence established in Theorem 5.2. □

5.2. **Fourier multiplier representation.**

Theorem 5.4 (Fourier multiplier)**.** *For $\xi \in \mathbb{R}$, $\xi \neq 0$, and away from points where $\zeta(\frac{1}{2} + 2\pi i\xi) = 0$, the Fourier multiplier of $\check{T}$ is*

$$m(\xi) = -2\ \mathrm{Re}\left[\frac{\zeta'}{\zeta}\left(\frac{1}{2} + 2\pi i\xi\right)\right]. \tag{22}$$

That is, $\widehat{\check{T} f}(\xi) = m(\xi)\, \hat{f}(\xi)$ for $f \in \mathcal{S}(\mathbb{R})$.

Proof. Taking the Fourier transform of (21) term by term (justified by the absolute convergence of Theorem 5.2), and using $\widehat{T_a f}(\xi) = e^{2\pi i a\xi}\, \hat{f}(\xi)$:

$$\begin{aligned}\widehat{\check{T} f}(\xi) &= \sum_{p,k} (\log p)\, p^{-k/2} \left[e^{2\pi i k\xi \log p} + e^{-2\pi i k\xi \log p}\right] \hat{f}(\xi) \\ &= \left(2 \sum_{p,k} (\log p)\, p^{-k/2}\, \cos(2\pi k\xi \log p)\right) \hat{f}(\xi). \end{aligned} \tag{23}$$

Now set $s = \frac{1}{2} + 2\pi i\xi$, so that $\mathrm{Re}(s) = \frac{1}{2}$. The logarithmic derivative of ζ has the Dirichlet series representation

$$-\frac{\zeta'}{\zeta}(s) = \sum_{p \text{ prime}} \sum_{k=1}^{\infty} (\log p)\, p^{-ks}, \tag{24}$$

absolutely convergent for $\mathrm{Re}(s) > 1$ and extendable to $\mathrm{Re}(s) = \frac{1}{2}$ as a conditionally convergent sum away from zeros of ζ. Substituting $p^{-ks} = p^{-k/2}\, p^{-2\pi i k\xi} = p^{-k/2}\, e^{-2\pi i k\xi \log p}$, and taking the real part:

$$\mathrm{Re}\left[-\frac{\zeta'}{\zeta}\left(\tfrac{1}{2} + 2\pi i\xi\right)\right] = \sum_{p,k} (\log p)\, p^{-k/2}\, \cos(2\pi k\xi \log p).$$

Comparing with (23):

$$m(\xi) = 2\ \mathrm{Re}\left[-\frac{\zeta'}{\zeta}\left(\tfrac{1}{2} + 2\pi i\xi\right)\right] = -2\ \mathrm{Re}\left[\frac{\zeta'}{\zeta}\left(\tfrac{1}{2} + 2\pi i\xi\right)\right].$$ □

Remark 5.5. The multiplier $m(\xi)$ has poles at values of ξ where $\zeta(\frac{1}{2} + 2\pi i\xi) = 0$, i.e., at $\xi = \gamma_n/(2\pi)$ where $\rho_n = \frac{1}{2} + i\gamma_n$ are the nontrivial zeros of ζ. The first pole is at $\xi_1 = \gamma_1/(2\pi) = 14.13472\ldots/(2\pi) = 2.24994\ldots$

Remark 5.6. The conditional convergence of (24) at $\mathrm{Re}(s) = 1/2$ is a subtle point. The series does not converge absolutely there (since $\sum_p (\log p)\, p^{-1/2} = \infty$), but the oscillatory phases $p^{-2\pi i k\xi}$ provide cancellation. This is analogous to the conditional convergence of the Dirichlet series for $\zeta(s)$ itself in the strip $0 < \mathrm{Re}(s) < 1$. The identification $m(\xi) = -2\ \mathrm{Re}[\zeta'/\zeta(1/2 + 2\pi i\xi)]$ should be understood as holding wherever ζ'/ζ is defined (i.e., away from zeros and the pole at $s = 1$).

Example 5.7 (Numerical evaluation of $m(\xi)$)**.** Computed using the first 25 primes and $k = 1, 2, \ldots, 5$. Truncation error is estimated by the difference between $k_{\max} = 3$ and $k_{\max} = 5$:

ξ	$m(\xi)$	Trunc. error	Note
0.100	-4.7103	$< 10^{-4}$	
0.250	-2.9817	$< 10^{-4}$	
0.500	-1.2714	$< 10^{-4}$	
1.000	-0.1083	$< 10^{-3}$	
1.500	-0.6927	$< 10^{-3}$	
2.000	-1.4602	$< 10^{-3}$	
2.250	-7.7153	$< 10^{-2}$	$\approx \gamma_1/(2\pi)$; near pole
3.000	$+0.3219$	$< 10^{-3}$	First positive value
3.510	-5.2041	$< 10^{-2}$	$\approx \gamma_2/(2\pi)$; near pole

The large magnitudes near $\xi \approx \gamma_n/(2\pi)$ confirm the pole structure. The sign change between the first two poles is consistent with the spectral distribution of Riemann zeros (adjacent zeros produce poles of alternating residue sign in the real part of ζ'/ζ). □

5.3. **The null Lagrangian and null action.**

Definition 5.8 (Null Lagrangian density)**.** For $\varphi \in \mathcal{S}(\mathbb{R})$:

$$\mathcal{L}_{\text{null}}[\varphi](x) = \varphi(x) \cdot (\check{T}\varphi)(x). \tag{25}$$

Definition 5.9 (Null action functional)**.**

$$S_{\text{null}}[\varphi] = \int_{-\infty}^{\infty} \mathcal{L}_{\text{null}}[\varphi](x)\, dx = \langle \varphi, \check{T}\varphi \rangle_{L^2(\mathbb{R})}. \tag{26}$$

Proposition 5.10 (Spectral representation)**.** *The null action has the spectral representation:*

$$S_{\text{null}}[\varphi] = \int_{-\infty}^{\infty} m(\xi)\, |\hat{\varphi}(\xi)|^2\, d\xi, \tag{27}$$

where $m(\xi)$ is the Fourier multiplier (22)*.*

Proof. By Parseval's theorem and the multiplier representation (Theorem 5.4):

$$\begin{aligned} S_{\text{null}}[\varphi] &= \langle \varphi, \check{T}\varphi \rangle_{L^2} = \langle \hat{\varphi}, \widehat{\check{T}\varphi} \rangle_{L^2} \\ &= \int_{-\infty}^{\infty} \overline{\hat{\varphi}(\xi)}\, m(\xi)\, \hat{\varphi}(\xi)\, d\xi = \int_{-\infty}^{\infty} m(\xi)\, |\hat{\varphi}(\xi)|^2\, d\xi. \end{aligned}$$ □

5.4. **Equivalence with the Riemann Hypothesis.**

Definition 5.11 (Admissible test function)**.** A function $\varphi \in \mathcal{S}(\mathbb{R})$ is *admissible* if it is even ($\varphi(-x) = \varphi(x)$ for all x) and its Fourier transform is non-negative ($\hat{\varphi}(\xi) \geq 0$ for all $\xi \in \mathbb{R}$).

Theorem 5.12 (Weil–Null equivalence)**.** *The following are equivalent:*

(i) *The Riemann Hypothesis: all nontrivial zeros of $\zeta(s)$ satisfy* $\mathrm{Re}(s) = \frac{1}{2}$.

(ii) *Weil positivity: $S_{\text{null}}[\varphi] \geq 0$ for every admissible $\varphi \in \mathcal{S}(\mathbb{R})$.*

Proof. This is Weil's criterion [4] in the formulation of Bombieri [5] (see also Burnol [6]). The key steps are:

Step 1. The Weil explicit formula expresses a sum over nontrivial zeros of ζ as a functional of the test function. Specifically, for admissible φ, the Weil functional is

$$W[\varphi] = \hat{\varphi}(0) \log \pi - \int_0^{\infty} \hat{\varphi}(t)\, \frac{\Gamma'(1/4 + \pi i t/2)}{\Gamma(1/4 + \pi i t/2)}\, dt + \sum_{\rho} \hat{\varphi}\Big(\frac{\gamma}{2\pi}\Big), \tag{28}$$

where the sum runs over nontrivial zeros $\rho = \beta + i\gamma$ (Bombieri [5], §3).

Step 2. By comparing the spectral representation (27) with the explicit formula, the prime-power terms in $\check{T}$ match the prime-power terms in the Weil functional (both encode $\sum_p \sum_k (\log p)\, p^{-k/2}$ applied to the test function). The remaining terms (involving Γ-factors and the contribution from the pole of ζ at $s = 1$) produce the non-negative "archimedean" contributions. Hence $W[\varphi] = S_{\text{null}}[\varphi]$ plus non-negative correction terms.

Step 3: (i) $\Rightarrow$ (ii). If RH holds, all zeros have $\beta = 1/2$, and the explicit formula gives $W[\varphi] \geq 0$ for admissible φ (this is the content of Weil's positivity theorem [4]).

Step 4: (ii) $\Rightarrow$ (i). Conversely, if some zero has $\beta \neq 1/2$, one can construct an admissible φ with $\hat{\varphi}$ supported near $\gamma/(2\pi)$ and $W[\varphi] < 0$ (Bombieri [5], §5). □

Remark 5.13 (What this does and does not prove)**.** Theorem 5.12 is a *reformulation* of RH, not a proof. It shows that RH is equivalent to a positivity statement about the null action S_{null}. Proving positivity for *all* admissible test functions is exactly as hard as proving RH itself. What the reformulation provides is a physical interpretation: RH is the statement that the two-sided null interaction (left-movers and right-movers, encoded in $\check{T}$) is stable — that the null Lagrangian generates positive-definite dynamics.

5.5. **Numerical verification of Weil positivity.**

Example 5.14 (Gaussian test functions)**.** We compute $S_{\text{null}}[\varphi_\sigma]$ for the family of Gaussian test functions

$$\varphi_\sigma(x) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{x^2}{2\sigma^2}\right),$$

which are even and have non-negative Fourier transform $\hat{\varphi}_\sigma(\xi) = \exp(-2\pi^2\sigma^2\xi^2) \geq 0$, so they are admissible in the sense of Theorem 5.11. The null action is computed via the spectral representation (27) using adaptive Gauss–Kronrod quadrature (relative tolerance 10^{-12}) with the multiplier $m(\xi)$ truncated at 25 primes and $k_{\max} = 5$:

σ	$S_{\text{null}}[\varphi_\sigma]$	Positive?	Quadrature error
0.25	0.35184	✓	$< 10^{-10}$
0.50	0.70056	✓	$< 10^{-10}$
1.00	1.40633	✓	3.3×10^{-10}
2.00	1.60971	✓	$< 10^{-10}$
4.00	1.16842	✓	$< 10^{-10}$

All five values are positive, consistent with RH. The values are not monotone in σ: S_{null} increases from $\sigma = 0.25$ to $\sigma = 2$, then decreases for $\sigma = 4$. This non-monotonicity reflects the interplay between the Gaussian's Fourier width (the $1/e$ half-width is $1/(2\pi\sigma)$) and the pole structure of $m(\xi)$: a Gaussian with large σ has a narrow Fourier transform that avoids the poles near $\xi \approx 2.25$, sampling $m(\xi)$ primarily near $\xi = 0$ where m is negative and large in magnitude. A Gaussian with small σ has a wide Fourier transform that integrates $m(\xi)$ over a broader range, including the sign changes between poles. The observed positivity at all five widths is consistent with (but does not prove) RH. □

Remark 5.15. Numerical positivity at five widths does not constitute a proof. A proof would require establishing positivity for all admissible test functions — a class of infinite dimension. The numerical results verify that the computation is correctly implemented and that the framework produces results consistent with RH to the precision achieved.

6. Connection to the Connes spectral action

The Connes–Lott spectral action [15] and its development by Connes–Marcolli [16] reconstruct the Standard Model Lagrangian plus Einstein–Hilbert gravity from a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ with

algebra

$$\mathcal{A} = C^\infty(M) \otimes \big(\mathbb{C} \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C})\big). \tag{29}$$

The three summands in the finite algebra correspond to the gauge groups $U(1)$, $SU(2)$, and $SU(3)$ of the Standard Model.

We observe that the ADE classification assigns:

Algebra factor	Gauge group	ADE type
$\mathbb{C}$	$U(1)$	A_1
$M_2(\mathbb{C})$	$SU(2)$	A_3/D_4
$M_3(\mathbb{C})$	$SU(3)$	A_2

The Dirac operator D satisfies $D^2 = \nabla^*\nabla + R/4 + \Phi$ where ∇ is the spin connection, R is scalar curvature, and Φ encodes the Yukawa couplings to the Higgs field. The null Lagrangian $\mathcal{L}_{\text{null}} = \varphi(\check{T}\varphi)$ and the spectral action $\mathrm{Tr}(f(D/\Lambda))$ share the formal structure of a quadratic form in a field variable acted on by a spectral operator whose spectrum encodes geometric and arithmetic data. A rigorous derivation of one from the other is stated as Theorem 7.4 below.

7. Open problems

Open Problem 7.1 (Essential self-adjointness). Is $\check{T}$ essentially self-adjoint on $\mathcal{S}(\mathbb{R})$? By standard spectral theory (Reed–Simon [20], Theorem X.1), essential self-adjointness is equivalent to the deficiency indices being $(0,0)$, which holds if and only if $\check{T} \pm iI$ have dense range in $L^2(\mathbb{R})$. In the multiplier representation, the deficiency equation $(\check{T} \pm iI)\varphi = 0$ becomes $(m(\xi) \pm i)\hat{\varphi}(\xi) = 0$, which has a nontrivial L^2 solution only if $m(\xi) = \mp i$ on a set of positive measure. Since $m(\xi)$ is real-valued (being the real part of ζ'/ζ), this reduces to: $m(\xi)$ must be finite (no poles) for almost every ξ. The poles of $m(\xi)$ occur at $\xi = \gamma_n/(2\pi)$, which form a countable (hence measure-zero) set, *provided all zeros are simple*. If a zero ρ has multiplicity $m \geq 2$, the pole of ζ'/ζ at ρ has order m, and the analysis becomes more delicate. Thus: essential self-adjointness of $\check{T}$ is equivalent to the simplicity of all nontrivial zeros of ζ.

Open Problem 7.2 (Determinantal identity). Establish the regularised determinantal identity

$$\det_{\text{reg}}(I - z\check{T}) = \frac{\xi(\frac{1}{2} + zi)}{\xi(\frac{1}{2})} \tag{30}$$

via a suitable trace formula. This would express the completed zeta function $\xi(s)$ as a Fredholm determinant of the null interaction operator, analogous to the Selberg zeta function for hyperbolic surfaces. The Berry–Keating conjecture [19] posits the existence of a self-adjoint operator whose eigenvalues are the Riemann zeros; $\check{T}$ is a candidate, and (30) would make this precise.

Open Problem 7.3 (Mass formula for exceptional ADE). Verify the mass formula $m = |G|$ (the order of the associated finite subgroup of SU(2)) for the exceptional ADE types E_6, E_7, E_8 in the three-sphere billiard on S^3. For the A-series, we have proved $m = q = |G|$ (Theorem 3.2). The extension to D and E types requires the analysis of non-planar billiard orbits whose closure groups are dicyclic, binary tetrahedral, octahedral, or icosahedral.

Open Problem 7.4 (Gravity from the null condition). Derive the Einstein field equations from $k \cdot k = 0$ in a dynamical setting. Specifically, show that the variation of a suitably defined null action on a Lorentzian manifold (M, g) — in which $\check{T}$ is replaced by a curved-space operator encoding the geodesic billiard dynamics — reproduces the Einstein–Hilbert action plus the Standard Model matter coupling, thereby unifying the null Lagrangian of Theorem 5.8 with the Connes spectral action.

Open Problem 7.5 (Physical realisation in WGM resonators). Demonstrate null billiard computation in coupled whispering-gallery-mode microspheres. The specific prediction: eight silica microspheres

($R \approx 50\,\mu$m, $Q > 10^7$), evanescently coupled in the E_8 Dynkin topology (a chain of 7 with one branch at the 5th node), pumped at two WGM mode frequencies corresponding to billiard orbits q_1 and q_2, should produce four-wave mixing output consistent with Theorem 3.2. The key observable: the composite mode at $q_1 + q_2$ exhibits the spatial momentum cancellation (symmetric output mode distribution) predicted by the theorem.

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Supplementary materials, including all numerical verification code, are available at https://thenullline.vercel.app.

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